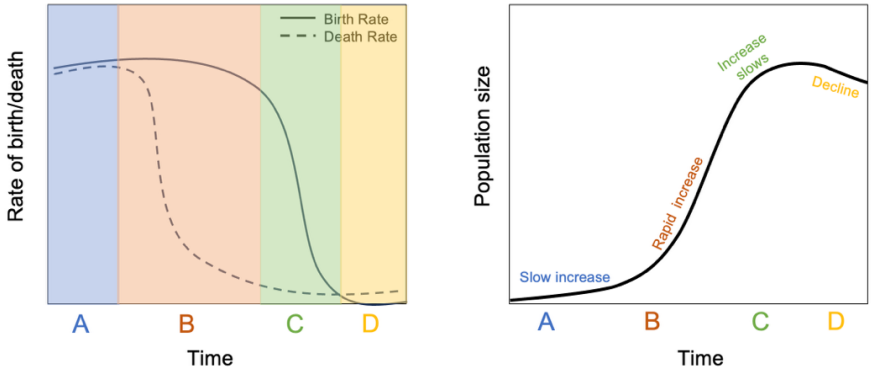
BIOLOGICAL SCIENCES 2B –GERHART
WINTER QUARTER 2026
MIDTERM #1 KEY

1. B
   Note that sexual reproduction is not a requirement of life – many organisms reproduce asexually. Maintaining particular internal conditions that differ from external conditions is a requirement of all living things (commonly called homeostasis) – examples we discussed in class were maintaining a stable body temperature and managing salt/water balance.
2. C
3. B
4. B
5. D
6.
   a. C
   b. C
7. *For full points, all four components of the graph below needed to be included: in the early part of the curve (time period A), growth is slow; as the death rate falls, the growth rate increases (time period B); as the birth rate also falls, the growth rate slows but is still positive (time period C) and at the end when death rates are higher than birth rates, population size decreases (time period D).*



8. Resting Blood Oxygen Levels: *the intended answer was F (more than 10 days) since resting blood oxygen levels never fully return to baseline values at sea level; however the question didn't fully define 'acclimation' in the sense of clarifying if I meant when baseline levels are achieved, or when the metric plateaus. Consequently, students could have interpreted "acclimation" as meaning "when levels return to baseline" (for which the answer was "more than 10 days") or interpreted "acclimation" as meaning "when changes in the metric plateau" (for which the answer could be "7-8 days"). Because the line was wobbly prior to fully plateauing, I also gave partial credit (1/2 points) for "4-5 days" and "5-6 days."*
   Cognitive Performance: B (*given the rapidity of acclimation in this metric, the plateau and return to baseline levels both occur at ~2-3 days, so the issue described above does not apply here)*
9. B
10. Possible explanations include: other prey species have important nutrients (not just energy) that the sea stars need; more sea star predators in regions where mussels are common; mussels are too rare to solely support the sea stars food intake needs, mussels have toxic compounds that the sea stars can't consume. *Answers that focused on sea stars being in competition with other predators were given partial credit – competition with other predators would mostly affect this decision only if it resulted in mussels being too rare to support sea stars (which makes this an indirect effect, with the primary issue being mussel scarcity). Answers that focused on relative abundance of mussels compared to other potential prey were not awarded points – being less common than other prey would not shift the predator choice so long as there are still enough mussels to support sea stars (the key point is absolute abundance not relative*

abundance). Answers that focused on energy return and investment were not awarded points – these answers restate the way OFT predicts prey choice, rather than describing a predator decision that might 'violate' OFT predictions. Lastly, an incorrect answer to question 9 did not affect the awarding of points for this question – regardless of which prey species OFT predicts, the same exceptions could all apply for sea stars making other decisions.

11. B and C
    On 'select all that apply' questions, partial credit is awarded in the following way: the total points for the question (in this case, 3) is split between all answer options (in this case 4, so each answer option is worth 0.75 points). Students are awarded points for 1) correctly selecting a correct answer option, and 2) correctly ***not*** *selecting an incorrect answer option.*
12. Location A: Temperate deciduous forest in the Northern Hemisphere
    Location B: Temperate grassland in the Southern Hemisphere
13. The loss of mycorrhizae means that below ground resources are now relatively more difficult for the trees to obtain (compared to above-ground resources). Consequently, the trees will invest more growth in below-ground tissues (roots) to better obtain these resources. Increased investment of growth in roots compared to shoots will increase the root:shoot ratio. *Note that the mycorrhizae are fungal organisms that are separate individuals from the tree, so the initial loss of mycorrhizae does not decrease the root:shoot ratio as the tree's mass has not changed.*
14.
    a. 3.788
       no calculations needed, the value was provided in the table. One point was deducted for typos or incorrect rounding
    b. 24.817/3.788 = 6.551 years
    c. (ln(3.788))/6.551 = 0.203
15.
    a) 150 = 350/(1+((350-279)/279)e^(-10r)) = 350/(1+(0.2545e^-10r))
       150(1+(0.2545e^-10r))=350
       1+(0.2545e^-10r) = 35/150 = 2.333
       0.2545e^-10r = 2.333-1 = 1.333
       e^-10r = 1.333/0.2545 = 5.239
       -10r(lne) = ln(5.239)
       -10r = 1.656
       -r = 0.1656
       r = -0.166
       Partial credit was awarded in the following way: correctly setting up the original equation was worth 2 points and correctly calculating the r value from that set up was worth 2 points. If only a single minor error was made in the calculation (mis-added 1+0.245 in the denominator, dropped the minus sign from r at some point, mis-located a decimal, etc), then 1 point was deducted. If multiple minor errors were made, or if errors were more serious than simple typos, then 2 points were deducted
    b) -0.166 = (lnR_0)/3
       3(-0.166) = lnR_0 = -0.4968
       e^(lnR_0) = R_0 = e^-0.4968
       R_0 = 0.608
       Partial credit was awarded in the following way: correctly setting up the original equation was

worth 2 points and correctly calculating R_0 from that set up was worth 2 points. If a student incorrectly calculated r in part A, but correctly used that value here to calculate R_0, no points were deducted. If only a single minor error was made in the calculation (rounding drift, dropped minus sign) then 1 point was deducted. If multiple minor errors were made, or if errors were more serious than simple types, then 2 points were deducted.

16. Estimate without knowing about molting is: 15/N = 9/23; 9N=23x15=345; N=345/9 = **38.333**
    Estimate with knowing about molting is: 10/N = 9/23; 9N=230; N=230/9 = **25.556**
    Your estimate will be **too high** because you think there are more tagged individuals in the population than there really are, which inflates your estimate of N. Knowing the true number of tagged individuals at the time of your second sampling (10, not 15) produces a lower N estimate.
    1 point was awarded for correctly calculating N without knowing about the molting, and 1 point was awarded for correctly calculating N if the molting was known. Two points were then awarded for fully and accurately explaining how the molting affected your estimate population size.